PertSpectra: Interpretable Matrix Factorization for Predicting Functional Impact of Genetic Perturbation Experiments

Seowon Chang[1], Anna Shcherbina[1], Tal Ashuach[1], Shahin Mohammadi[1], Stephanie See[2], Ninad Ranadive[2], Emily Fox[1], Navpreet Ranu[2]
[1] Data Science and Machine Learning, [2]Functional Genomics
insitro, South San Francisco, CA, USA

Abstract

In drug discovery, measuring the effects of genetic perturbations is a powerful tool for studying unknown disease mechanisms, but biological interpretation of these effects, especially with the advent of screens involving combinatorial perturbations, remains challenging. To address limitations in current methodology we introduce PertSpectra, a guided triple matrix factorization that incorporates perturbation information and regularizes the model using a known gene-gene interaction graph prior to generate sparse, biologically relevant latent factors that capture perturbational effects. We evaluate PertSpectra on three single-cell RNA-seq datasets with both single and combinatorial genetic perturbations, measuring latent space interpretability, predictive ability on unseen combinations of observed perturbations, and stratification of functionally similar perturbations. We show that PertSpectra provides an integrated modeling approach to understanding combinatorial perturbation data in the context of drug discovery.

Introduction

- Perturb-seq has emerged as a powerful tool in studying how individual genetic perturbations affect cellular phenotype.
- Accurately identifying perturbation signals, which are weaker than background signal and noise, and disentangling multiple perturbations pose computational challenges for analyzing these datasets.

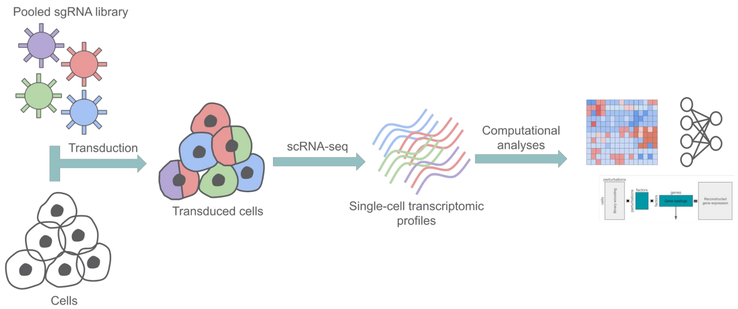

Figure 1. Perturb-seq experiment and analysis example workflow.

- Current computational models are either challenging to interpret or do not incorporate biological priors into the model design, limiting identifiability with the underlying biology.
  - GSFA[4] is a guided matrix factorization for perturbation experiments that explicitly includes perturbation labels in the model.
  - Spectra[2] is a scRNA-seq matrix factorization model for incorporating biological priors into model training.
- We propose PertSpectra, designed to correctly predict single and combinatorially perturbed expression profiles and learn biologically interpretable latent factors that yield insights into the functional impact of perturbations.

PertSpectra

PertSpectra is a guided triplet matrix factorization model that incorporates both perturbation information from the experiment as well as biological knowledge graphs:

$$\mathbb{E}[\mathbf{X_{ij}}] = (\nu_j + \delta)\boldsymbol{H}_i^\intercal \boldsymbol{P}\boldsymbol{W}_j$$

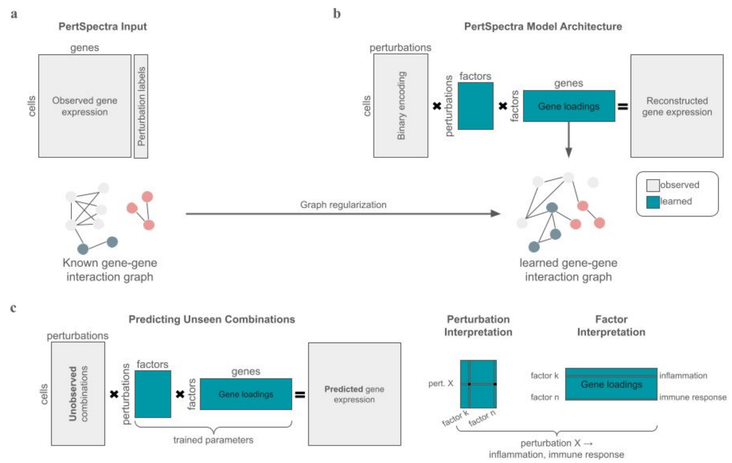

Figure 2. a. Input for PertSpectra, b. Illustration of PertSpectra model, c. Downstream applications of a trained PertSpectra model

The overall loss function is given by

$$\mathcal{L}(\boldsymbol{\Theta}) = \lambda \mathcal{L}_{recon}(\boldsymbol{\Theta}) + \mathcal{L}_{graph}(\boldsymbol{\Theta}).$$

Where the reconstruction loss is

$$\mathcal{L}_{recon}(\boldsymbol{\Theta}) = \boldsymbol{\eta}\big[-\mathbf{X}\log(\hat{\mathbf{X}}(\boldsymbol{\Theta})) + \hat{\mathbf{X}}(\boldsymbol{\Theta})\big]$$

And the graph regularization loss is

$$\mathcal{L}_{graph}(\boldsymbol{\Theta}) = -\mathbf{A}_{ij}\log(\mathbb{P}[\mathbf{A}_{ij}=1]) - (1-\mathbf{A}_{ij})\log(1-\mathbb{P}[\mathbf{A}_{ij}=1])$$

This factorization gives us (1) biologically interpretable gene loadings and (2) an association of the latent factors to perturbations, rather than individual cells.

Evaluation

Dataset	Type	# Cells	# Perts
In-house	Combinatorial	65648	182
Norman	Combinatorial	101484	235
Replogle	Single	105943	517

Table 1. Evaluation datasets

- We evaluated PertSpectra on 3 perturb-seq datasets[2,3].
- We demonstrated PertSpectra's ability to generate coherent, biologically interpretable latent factors while maintaining or improving performance on predictive tasks compared to benchmark models.

Results

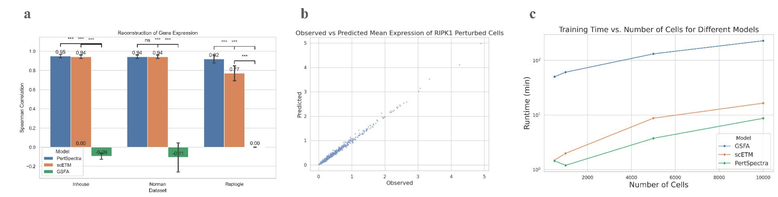

Figure 3. a. Spearman correlation between predicted and observed mean gene expression by perturbation, b. Predicted vs observed mean expression for the perturbation RIPK1 (inhouse), c. Training runtime vs. number of cells across different models

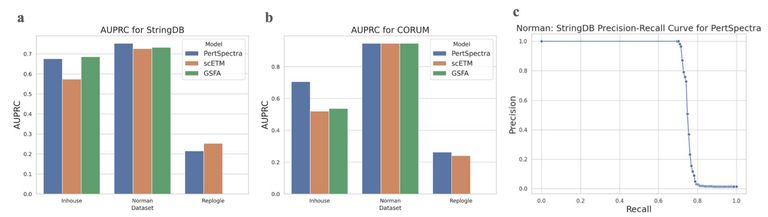

Figure 4. a. Area under the Precision-recall curve (AUPRC) for StringDB known relationships, b. AUPRC for CORUM known relationships, c. Precision-recall curve for PertSpectra on the Norman dataset

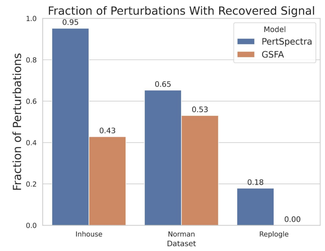

Figure 5. Recovery of known biological functions from hypergeometric test interpretability analysis

- We performed GSEA on the gene loadings to assign processes to each latent factor.
- We can associate these processes to perturbations using the factorization.
- Performing a hypergeometric test between the model predicted associations and known biology reveals the model's recall ability.

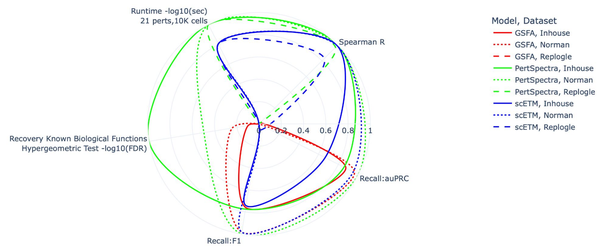

Figure 6. Radar plot summarizing model performance across metrics/downstream analyses

References

1. Kunes, R. et al. Supervised discovery of interpretable gene programs from single-cell data. Nature Biotechnology, 42:1084—1095, 2024.
2. Norman, T. et al. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. Science, 365(6455):786–793, 2019.
3. Replogle, J. et al. Mapping information-rich genotype- phenotype landscapes with genome-scale perturb-seq. Cell, 85(14):2559–2575, 2022.
4. Zhao, Y., Luo, K., Liang, L., Chen, M., and He, X. A new bayesian factor analysis method improves detection of genes and biological processes affected by perturbations in single-cell crispr screening. Nature Methods, 20:1693—1703, 2023.